



Digital Integrated Circuit Design

Lecture-10: Low Power Design

Chih-Feng Wu (吳志峰), PhD.

*Department of Electronics Engineering, Chang Gung University,
Taoyuan, Taiwan (ROC)*

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Outline

- ☒ Low-Power Design
- ☐ Some DSP Tricks

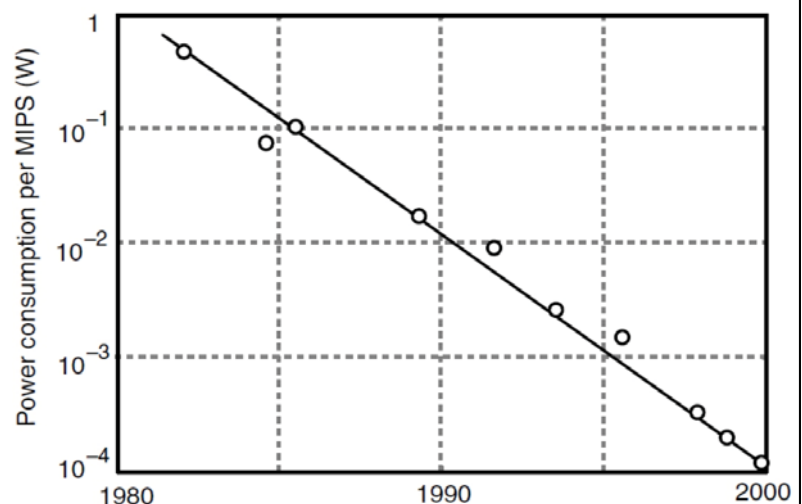
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Need for Low-Power Design

- ❑ Portable and wearable electronic devices
 - Lithium-ion battery: 0.2 watt-hr per gram of weight
 - Practical battery weight: < 500 g (< 50 g for wearable device)
 - Total power $\cong$ 5-10 watt for a day work between recharges
- ❑ Modern high-performance microprocessors use 100s watts
 - Power $\propto$ die area $\times$ clock frequency
 - Difficult for cooling of micro-processor, but still manageable
 - A BIG challenge for cooling of MPPs and server farms
- ❑ New battery technologies cannot keep pace with demand.
 - Demand for more speed and functionality (multimedia, etc.)

Need for Low-Power Design (cont'd)

- ❑ Power consumption trend for each MIPS (million instructions per second) of computational performance in DSP chips
 - MIPS reduction of 10% per 5-year
- ❑ Key factors
 - Power management method
 - Low supply voltage



Sources of Power Consumption

❑ Power dissipation in CMOS digital circuits

- Static: Leakage current in imperfect switches (< 10%)
- Dynamic: Due to charging and dis-charging of parasitic capacitance

$$P_{avg} \cong \alpha f C_L V_{dd}^2$$

“activity” data rate (clock frequency) Capacitance Square of voltage

❑ For a given data rate f , there are 3 ways to reduce the power requirements:

- 1. Using a lower supply voltage V_{dd}
- 2. Reducing the parasitic capacitance C_L
- 3. Lowering the switching activity α

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Sources of Power Consumption (cont'd)

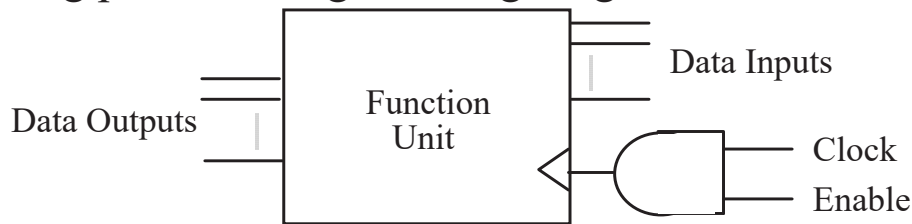
❑ **Example:** A 32-bit off-chip bus operates at 5 V and 100 MHz and drives a capacitance of 30 pF per bit. If random values were put on the bus in every cycle, we would have $\alpha = 0.5$. To account for data correlation and idle bus cycles, assume $\alpha = 0.2$. Please find its power consumption.

■ Sol.:

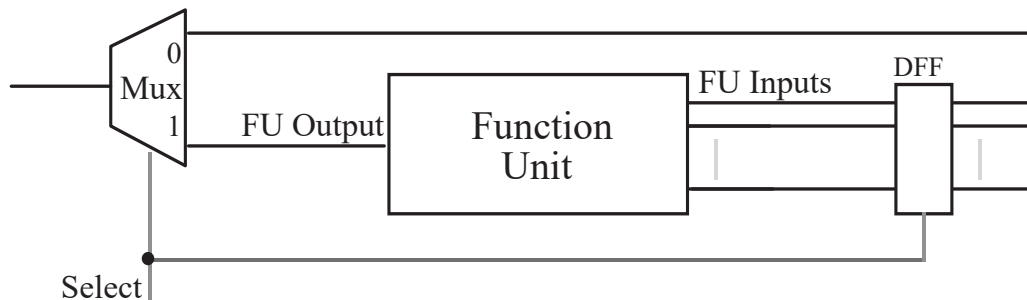
$$P_{avg} \cong \alpha f C_L V_{dd}^2 = 0.2 \times 100 \text{ MHz} \times (32 \times 30 \text{ pF}) \times 5^2 = 0.48 \text{ W}$$

Reduction of Power Waste

❑ Saving power through clock gating



❑ Saving power via guarded evaluation



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Reduction of the Number of Arithmetic Operations

❑ Useful techniques to reduction of the number of arithmetic operations

- $P_{(\text{shifting+add})} < P_{(\text{multiplier})}$
- $P_{[a(b+c)]} < P_{[ab+ac]}$

❑ Low-power arithmetic for CPLX multiplication

$$(a + jb)(c + jd) = (ac - bd) + j(ad + bc)$$

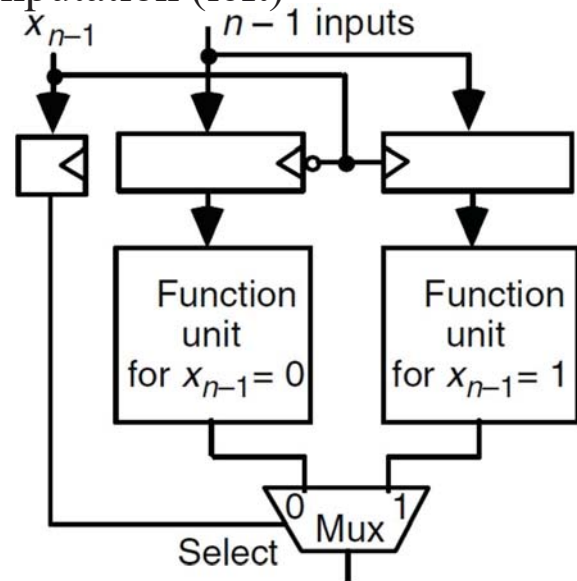
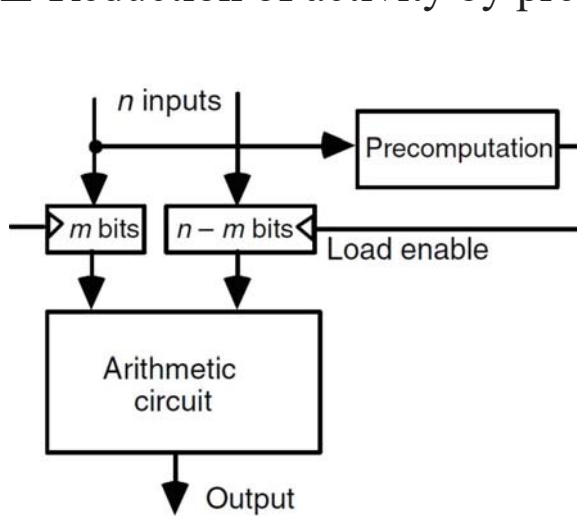
$$(a + jb)(c + jd) = [c(a + b) - b(c + d)] + j[c(a + b) - a(c - d)]$$

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Reduction of Activity α

- Reduction of activity by precomputation (left)



- Reduction of activity via Shannon expansion (right)

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Transformations and Trade-offs

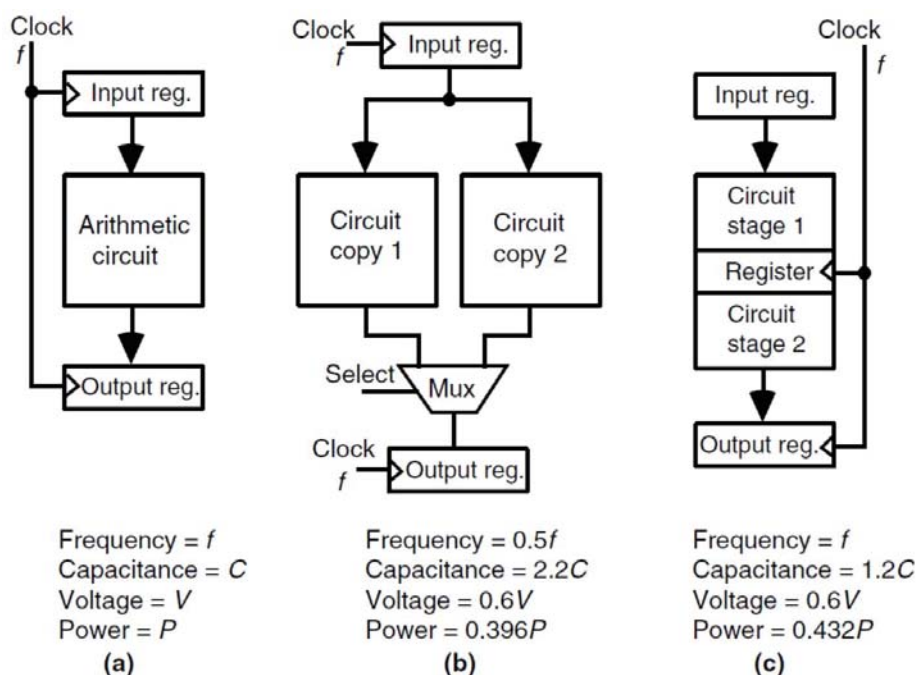
- Parallelism and pipelining
- Two-way parallel circuit
 - Two-stage pipelined circuit
- Also, we can “**Parallelism and pipelining**” to reduce the power consumption.

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Transformations and Trade-offs

Reduction of power via parallelism or pipelining



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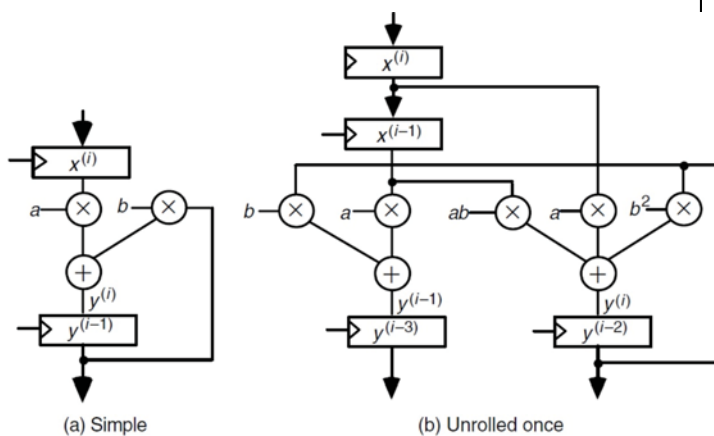
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Unrolling of Iterative Computations

Realization of a first-order IIR filter

Simple:

- $y_n = ax_n + by_{n-1}$
- 2-input adder
- Operating frequency f_{CLK}



Unrolling

- $y_n = ax_n + by_{n-1}$
- $y_{n+1} = ax_{n+1} + abx_n + b^2y_{n-1}$
- 2-input adder and 3-input adder
- Operating frequency f_{CLK}

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Retiming for Power Efficiency

Retiming

- Reduce power
- Increase throughput

Example: 4th order FIR filter

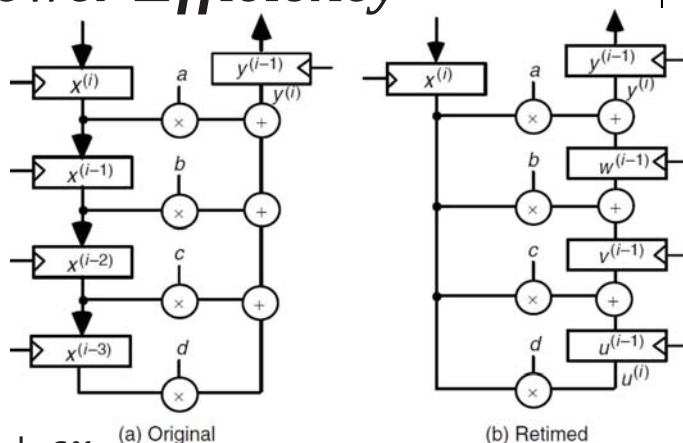
■ $y_n = ax_n + bx_{n-1} + cx_{n-2} + cx_{n-3}$

■ (a)

- Delay time = 1-MUL + 3-Add (
- Reduced by CSA + CPA (two-level adder tree)
- Less regular and more difficult to expand in a modular design (diadv.)

■ (b)

- A higher operating frequency with a given supply voltage
- A lower supply voltage for a desired throughput



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Outline

Low-Power Design

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High Speed Magnitude Approximation

- Magnitude Approximation for CPLX signal x_n

$$|x_n| = \sqrt{x_{n,R}^2 + x_{n,I}^2}$$

- Using $\alpha\mathbf{Max} + \beta\mathbf{Min}$ approximation to find $|x_n|$

- Also called alpha max plus beta max

$$|x_n| \approx \alpha\mathbf{Max} + \beta\mathbf{Min}$$

$(\alpha, \beta): (1, 1/2), (1, 3/8), (1, 4), \text{ or } (15/16, 15/32)$

$(\alpha, \beta): (1, 0.4), \text{ or } (0.9486, 0.39293)$

$$\mathbf{Max} = \max(|x_{n,R}|, |x_{n,I}|) \text{ and } \mathbf{Min} = \min(|x_{n,R}|, |x_{n,I}|)$$

High Speed Magnitude Approximation

- More accurate of $\alpha\mathbf{Max} + \beta\mathbf{Min}$ approximation

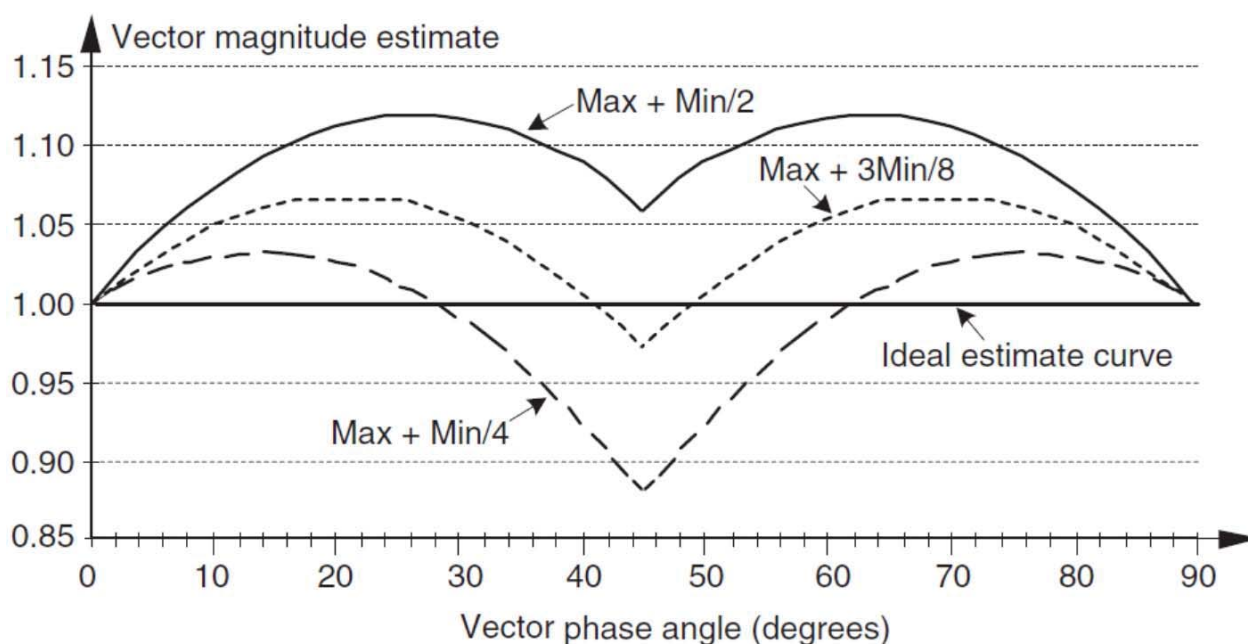
- Using multiply /shift and compare operation

$$|x_n| = \begin{cases} \mathbf{Max} + \frac{1}{8}\mathbf{Min}, & \text{if } \mathbf{Min} < \frac{3}{8}\mathbf{Max} \\ \frac{27}{32}\mathbf{Max} + \frac{19}{16}\mathbf{Min}, & \text{if } \mathbf{Min} \geq \frac{3}{8}\mathbf{Max} \end{cases}$$

- Using floating-point arithmetic operation

$$|x_n| = \begin{cases} 0.99\mathbf{Max} + 0.197\mathbf{Min}, & \text{if } \mathbf{Min} < 0.4142135\mathbf{Max} \\ 0.84\mathbf{Max} + 0.561\mathbf{Min}, & \text{if } \mathbf{Min} \geq 0.4142135\mathbf{Max} \end{cases}$$

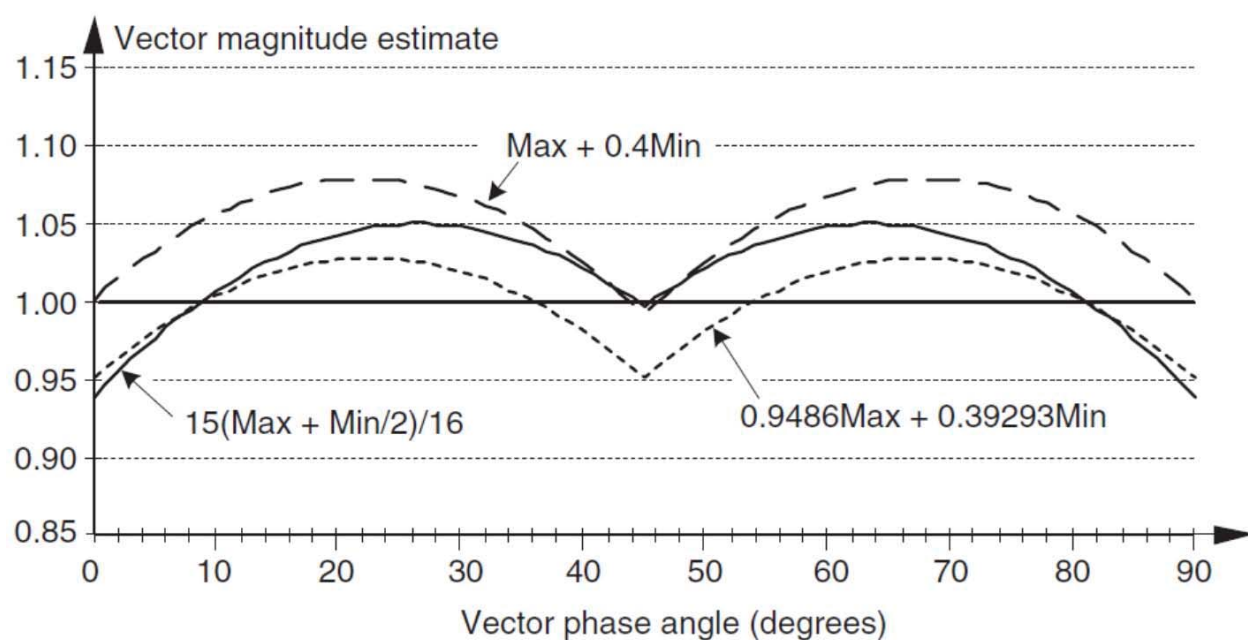
High Speed Magnitude Approximation(cont'd)



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High Speed Magnitude Approximation(cont'd)

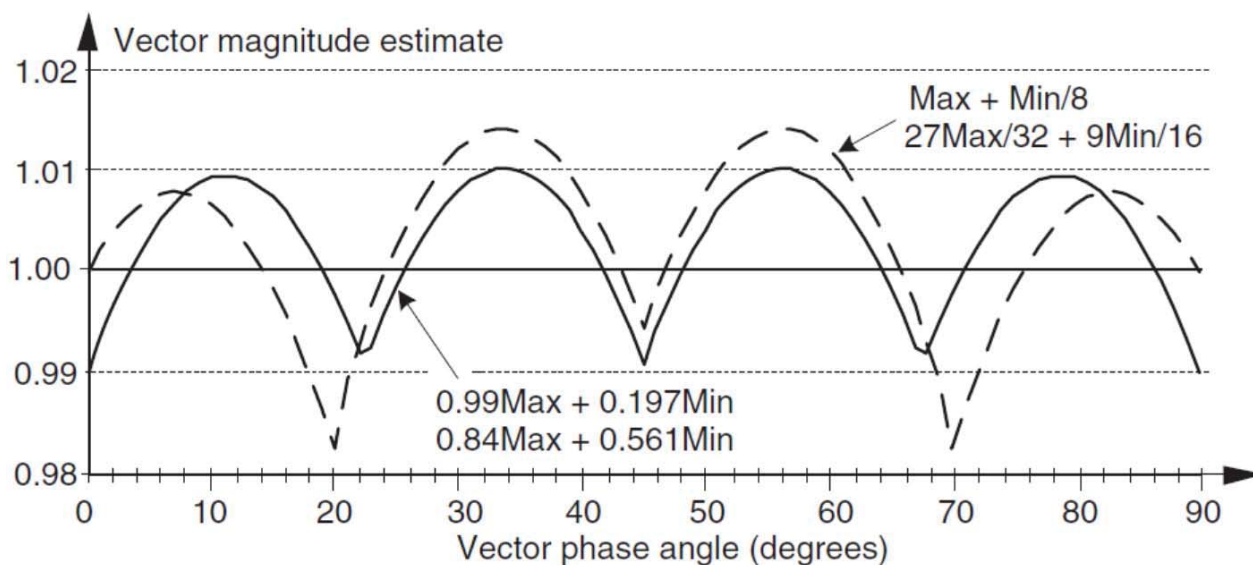


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High Speed Magnitude Approximation(cont'd)

□ Addition algorithm performance



Performance Comparison

Algorithm	Maximum error (%)	Maximum error (dB)	Mean squared error ($\times 10^{-3}$)
Max + Min/2	11.8%	0.97 dB	8.4851
Max + Min/4	11.6%	0.95 dB	1.7302
Max + 3Min/8	6.8%	0.57 dB	2.2682
15(Max + Min/2)/16	6.3%	0.53 dB	1.2412
0.9486Max + 0.39293Min	5.1%	0.43 dB	0.5773
Max + 0.4Min	7.7%	0.64 dB	3.0573
Max + Min/8, 27Max/32 + 19Min/16	1.8%	0.16 dB	0.0744
0.99Max + 0.197Min, 0.84Max + 0.561Min	1.0%	0.09 dB	0.0456